Savings in Relearning After Machine Unlearning: A Study of Learning Traces in Language Models

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Abstract

Machine unlearning tries to make a language model forget specific information without damaging the rest of what the model can do. One problem is that a model may stop giving the right answer without fully losing the effects of having learned that information before. This proposal borrows the idea of savings in relearning from human memory to test that possibility. An open-weight language model will first learn newly generated facts, go through either a weaker or more robust unlearning method, receive different amounts of unrelated training, and then learn the same facts again. The main measure will be how many target-data tokens the model needs before it reaches a fixed learning criterion. The most important comparison is between models that learned the target facts before and matched controls that never learned them. We will also check answer likelihoods, paraphrases, retained ability, abstention, and hallucination so that temporary hiding of an answer is not confused with stronger forgetting. Overall, the study asks whether earlier learning makes later relearning cheaper after unlearning, whether this advantage fades with more intervening training, and how far this behavior can reasonably be compared with human memory.

1 Introduction

We humans often relearn previously learned material much faster than we learned it the first time, even when we can no longer recall it directly. This savings-in-relearning phenomenon raises a question in the machine world: after a language model learns a set of facts and then "forgets" them, does its earlier learning reduce the training needed to learn the same facts again? Going one step further, we would also like to ask whether any such effect becomes weaker as the model receives more unrelated training after unlearning and before relearning the facts.

These questions matter for cognitive modeling because a savings-like pattern could give us a behavioral similarity between human memory and machine memory. Note that this is not to say that humans and machines use the same memory mechanisms. Instead, we would like to compare only what they do under a similar relearning setup. In this proposal, we ask whether earlier factual learning changes later relearning cost after robust unlearning, and whether the pattern we observe looks at all like savings in human memory.

2 Background

2.1 How do we measure savings in human memory?

Ebbinghaus's forgetting work is often remembered for showing that memory becomes worse over time, but an important measure in the original paradigm is savings. Savings asks how much learning effort is saved the second time because the material was learned once before (Murre and Dros, 2015). In simple form, the measure is:

Savings = (original learning effort - relearning effort) / original learning effort

A positive savings score means that earlier learning made the second round of learning easier, even if direct recall had already failed. This idea is useful for our project because an unlearned language model may stop producing the correct answer while still keeping some effect of the earlier training that changes what happens during later optimization.

There is one important difference between the human-memory setup and the study proposed here. Ebbinghaus varied the amount of time between learning and relearning, while our study changes the model through active parameter updates from unlearning and unrelated fine-tuning. In other words, intervening training is a measure of interference or model change, not a machine version of elapsed human time. For this reason, we treat the project as a test of savings after machine unlearning, not as a test of whether the exact Ebbinghaus forgetting curve also applies to language models.

2.2 Can we compare human memory and machine memory?

Recent cognitive science work suggests that it can still be useful to compare human and machine memory at the level of behavior, as long as we do not jump from similar behavior to similar internal mechanisms. Yu, Ku, and Griffiths (2026), for example, show that a finite-memory agent can develop human-like forgetting curves because forgetting can be useful under resource limits. The agent does not need to copy a biological human memory system to produce a similar pattern. Sanna et al. (2026) also discuss gradual changes in how earlier evidence remains available during human belief updating. Together, these studies give us a reason to look for a graded relearning advantage while keeping the explanation open.

Language models have also been tested on classic human-memory tasks. Wang et al. (2026) report that language models often remember more reliably than people, while prompting and external compression can make their behavior look more human-like. For our purposes, this suggests two things. First, human-memory effects can be useful behavioral benchmarks for language models. Second, matching a human-looking curve does not prove that the same internal process produced it. The strongest conclusion we would want to make here is simply that earlier training leaves some lasting effect that lowers later learning cost.

2.3 How do we make machines "forget" something?

This is the goal of machine unlearning. Machine unlearning tries to remove specific information, usually called the forget set, from a pretrained model while keeping the model's broader abilities through a retain set (Singh et al., 2025). Singh et al. (2025) introduce JensUn, which uses Jensen-Shannon divergence objectives for both forget and retain data and reports a useful trade-off between forgetting and retained performance. They also introduce the Lesser Known Facts (LKF) dataset and argue that forgetting should be tested with paraphrases and different input forms instead of only one original question. This matters because one wrong answer does not necessarily mean that the fact is gone. A model may still recover it when the same question is asked in another way or when extra context is added.

At the same time, unlearning does not always remove everything we want it to remove. Hu et al. (2024) show that knowledge targeted by several unlearning methods can return after a relatively small amount of additional fine-tuning, sometimes even when that training is only loosely related to the forgotten information. This suggests that some unlearning methods may mainly suppress access to knowledge rather than remove every effect of the earlier training. Still, fast recovery is not automatically the same as psychological savings. It could come from leftover correct-answer probability, repeated surface forms, or some other simple advantage. This is why we need a controlled first-learning stage and a control model that never learned the target facts in the first place.

2.4 How do we measure training effort?

For this study, training effort will mainly be measured by the cumulative number of target-data tokens needed to reach a fixed accuracy criterion. We use target tokens instead of only counting optimizer steps because one step can represent different amounts of actual training depending on batch size, sequence length, gradient accumulation, or how much target data is mixed into the batch. The same learning criterion will be used during first learning and relearning, which lets us ask directly whether the second learning stage is cheaper than the first (Murre and Dros, 2015).

2.5 What problems may occur?

One problem is contamination. If we use a public factual benchmark, some of those facts may already have appeared in the model's pretraining data, which would make the true first-learning cost impossible to know. A model failing a baseline question helps, but it does not prove that the fact is absent from its parameters. Exact string matching is also not enough because a paraphrase can carry the same information without using the same words (Yang et al., 2023). For this reason, the main dataset will use fresh, procedural entity-attribute mappings that could not have appeared in pretraining. LKF will be used only as a secondary test.

A second problem is how we judge open-ended answers. Singh et al. (2025) show that an LLM judge can recognize semantically correct answers better than simple word overlap, but LLM judges can also be biased or inconsistent (Shi et al., 2025). We will therefore use exact or alias-based scoring whenever possible, use an LLM judge only for ambiguous natural-language cases, hide the experimental condition from that judge, and compare some of its decisions with human annotations.

The past work reviewed here gives us many pieces of the puzzle, but it does not yet combine controlled first learning, weight-level unlearning, direct relearning of the same facts, a matched model learning those facts for the first time at the final stage, and graded target-free training in between. Putting these pieces together is the main contribution of the proposed study.

3 Proposed Study

The broad question is simple: after a language model has learned a new set of facts and those facts have been robustly suppressed, is the model still cheaper to train on the same facts than a matched model with no earlier history of learning them? We would also like to know whether any such advantage changes as more unrelated training is placed between unlearning and relearning.

3.1 Research questions and hypotheses

Hypothesis 1: Savings after robust unlearning. If earlier learning leaves a target-specific effect behind, then models that learned target set A before should reach the relearning criterion with fewer target tokens than matched controls that never learned A, even after forgetting has been robustly verified.

Hypothesis 2: Effects of unlearning strength. If weaker unlearning leaves more of the original learning behind, savings should be larger after the weaker unlearning treatment than after JensUn.

Hypothesis 3: Effects of intervening training. If unrelated training gradually weakens the remaining effect of earlier learning, the savings advantage should become smaller as the amount of intervening training increases.

Hypothesis 4: Target specificity. Earlier learning should make old target facts easier to reacquire, but it should not create the same advantage for a matched new fact set C. This lets us distinguish a memory-like target effect from a general change in how easy the model is to train.

Null hypothesis. After robust unlearning, the relearning effort may be practically the same as the effort of the never-learned control. We will define a preregistered

equivalence range so that a null result is not based only on a nonsignificant test.

3.2 Methods

3.2.1 Design and experimental units

There will be no human participants in this experiment. The experimental units will be independently trained copies of an open-weight language model, where we can directly access the weights, optimizer states, token probabilities, and checkpoints. We will choose a model around the 1-3 billion parameter range so that the full train-unlearn-intervene-relearn sequence can be repeated across conditions and random seeds without making the project unrealistic to run.

3.2.2 Materials and datasets

The main materials will be newly generated procedural facts. The dataset will contain 100 target facts, 100 matched novel-control facts, and 400 retain questions. Each fact will have separate prompt families for first learning, relearning, validation, and final testing. Validation and final-test prompts will never be used for gradient updates. Any LLM-generated paraphrases used for evaluation will also be checked for answer leakage, ambiguity, and whether they preserve the intended meaning.

LKF from Singh et al. (2025) will be included as a secondary dataset. It contains 100 forget questions and 400 retain questions across five niche historical topics, together with a paraphrase-based evaluation setup. We will include only items that the chosen base model fails across several prompt forms and that pass contamination screening. Because LKF is public, these results will be treated as an external validity check rather than proof that the model had never seen the information before.

3.2.3 Experimental branches and controls

The core design crosses two unlearning procedures with four amounts of intervening training. The weaker procedure will be a conventional baseline that increases loss on the forget set while trying to preserve retained performance. The more robust procedure will be JensUn. Importantly, we will define "strong" unlearning by what the model actually does, not just by the name of the method: target performance should move close to the never-learned or rollback reference while retained utility stays inside a preregistered margin.

Table 1

Experimental branches

Branch	Sequence	Purpose
Prior-learning	Learn target A -> unlearn A -> intervene -> relearn A	Measures the proposed savings effect
Naive-unlearning control	Learn matched set B -> apply the same unlearning procedure to A -> intervene -> learn A	Controls task practice caused by unlearning
Sham-unlearning control	Learn A -> receive matched retain-only or nontarget updates -> intervene -> relearn A	Estimates the upper bound when target learning is not actively removed
Rollback control	Restore the checkpoint from before learning A -> intervene -> learn A	Provides an exact no-target-history reference
Novel-fact control	After each branch, learn matched new set C	Tests whether effects are target-specific or reflect general learning changes

Note. A, B, and C denote matched fact sets.

All branches will receive matched practice on the same task format. The naive-unlearning branch is important because the unlearning objective itself could change the model's general learning rate even if A was never learned. The rollback branch gives us the cleanest reference for a model with no target-learning history.

3.2.4 Procedure

First of all, we will screen the base model on every target mapping using direct questions, sentence completions, and contrastive prompts. If the model already shows an unexpectedly high likelihood of producing the correct answer, or can generate the answer accurately before training, that item will be replaced. This is to make sure that the target facts are genuinely unfamiliar to the model before the experiment begins.

During the first learning stage, the prior-learning and sham branches will be fine-tuned on target set A, while the naive-unlearning branch will receive the same format of training on a matched set B. Rather than training each branch for a fixed number of steps, we will continue until a held-out validation criterion is reached. A candidate criterion is at least 90% factual accuracy together with a mean length-normalized log-likelihood margin of at least 1.0 nat for the correct answer over matched decoys at two consecutive evaluations. A pilot will be used to make sure this criterion is neither so easy that every model reaches it immediately nor so difficult that it is unrealistic to reach.

Next, we will apply the assigned unlearning procedure. Because a model may look as if it has forgotten something under one prompt while still revealing it under another, forgetting will be tested in several ways: original-form

questions, held-out paraphrases, sentence completions, augmented prompts, and correct-answer likelihood margins. We will also check retain accuracy, response quality, and general instruction-following ability so that target forgetting is not achieved by simply damaging the whole model. Hyperparameters will be selected on a development attack set, while a separate final attack set will stay untouched until the main evaluation.

After unlearning, the models will receive 0, 50,000, 250,000, or 1,250,000 tokens of target-free intervening fine-tuning. These doses are log-spaced because we would like to see whether any savings effect changes gradually as more unrelated training is added. The intervening corpora will be sampled independently for each run and screened for target entity names, answer aliases, semantic overlap, and statements that logically imply a target mapping. We will also check target likelihood and retained utility at checkpoints during this stage. Note that this manipulation represents training interference or model change, not the simple passage of time.

Before the final learning stage, we will reinitialize the optimizer, learning-rate scheduler, gradient scaler, and all optimizer moments. This matters because we want any difference in relearning effort to come from the model's learning history, not from leftover optimizer state. Batch size, target-to-retain ratio, trainable parameter subset, learning rate, and evaluation frequency will be kept the same across first learning and relearning. The previously trained branches will then relearn target set A using the counterbalanced prompt family, while matched controls will learn A for the first time. Every branch will also learn novel set C so that we can check whether any advantage is specific to old facts or simply reflects a broader change in plasticity.

3.2.5 Outcome measures

The main question here is how much training the model needs before it can learn the target facts again. Our primary measure will therefore be the cumulative number of supervised target tokens processed before the model reaches the held-out acquisition criterion. Target tokens include only examples tied to the target facts, not retain examples or unrelated intervention data. We prefer this over raw optimizer steps because steps can hide differences in batch size, sequence length, gradient accumulation, or the proportion of target data in each batch. Optimizer steps, total tokens, estimated floating-point operations, and wall-clock time will still be reported as secondary descriptive measures.

We will calculate two savings scores. The first is the traditional within-model score:

$$S_{within} = 1 - (E_{relearning} / E_{first\text{-}learning})$$

This score simply asks whether the same model needed less effort to learn the target facts the second time than it needed during first learning.

More importantly, we will also calculate a control-based savings score:

$$S_{control} = 1 - (E_{prior\text{-}history} / E_{never\text{-}learned})$$

A positive S_control would mean that earlier target learning reduced final learning effort compared with a matched model that never learned the targets, even after accounting for task practice, intervention history, and exposure to the unlearning procedure. We will also report raw effort differences and log effort ratios because ratio scores can become unstable when first learning is very fast. If a run does not reach criterion within a preregistered maximum budget, it will be treated as right-censored instead of being assigned the maximum value as though that effort had actually been observed.

Secondary outcomes will include area under the validation-error curve, acquisition slope, starting correct-answer likelihood, final test accuracy, likelihood margin over decoys, retain utility, and the proportion of runs that reach criterion. These extra measures matter because faster relearning could mean different things. A model might simply start closer to the right answer, or it might truly learn at a steeper rate. Looking at both starting point and learning curve should help us separate those cases.

3.2.6 Contamination, hallucination, and evaluation safeguards

Because our main question is whether something from earlier learning remains after unlearning, we need to make sure that apparent savings is not just contamination. We will use fresh procedural mappings for the main dataset, run exact, entity-level, and semantic checks across training, validation, final-test, and intervening corpora, deduplicate and counterbalance prompt families, and freeze scripts, seeds, model versions, and generated mappings before the main study.

We would also like to separate genuine forgetting from behaviors that only look like forgetting. Generated answers will be coded as correct, partially correct, abstaining or refusing, incorrect but related, or fabricated and unrelated. Hallucination will be reported separately from abstention because a model that says "I do not know" is behaving differently from a model that confidently invents an answer. Greedy decoding will be the main generation setting, every query will start with a fresh context, and probability-based measures will be reported along with generations so that learning a refusal phrase is not mistaken for removing the correct-answer distribution.

For the procedural facts, normalized exact matching and a registered alias list should handle most judgments. For LKF, an LLM judge will be used only when the output is genuinely ambiguous. The judge will see the question, gold answer, and candidate response but not the experimental condition or phase. Two human annotators will independently label a stratified sample, disagreements will be resolved, and we will report the judge's sensitivity, specificity, and agreement. We will also report worst-case success across paraphrases next to average accuracy and compare both with the identically evaluated never-learned baseline. This prevents a larger number of prompts from making recovery look better just through chance or judge error.

3.2.7 Analyses

For the main analysis, we will use a hierarchical accelerated failure-time model for the number of target tokens needed to reach criterion. The main reason for this choice is practical: some runs may not reach criterion within the allowed training budget, and this model can keep those runs as right-censored observations instead of pretending that we know their exact learning cost. Fixed effects will include prior-learning history, unlearning procedure, the log of intervening-training dose, and their interactions. Runs will be blocked by fact-set identity, with a random intercept for fact-set seed when the same generated set is reused across training seeds. In simplified form:

$$\log(E) = \beta_0 + \beta_1(\text{History}) + \beta_2(\text{Unlearning}) + \beta_3 \log(1 + \text{Dose})$$

$$+ \beta_4(\text{History} \times \text{Unlearning}) + \beta_5(\text{History} \times \text{Dose}) + u_{\text{fact-set}} + \text{error}$$

Each hypothesis maps onto a particular comparison. H1 is supported if prior-history models need less effort than the naive-unlearning control after robust forgetting has been verified. H2 is supported by a history-by-unlearning interaction, and H3 by a history-by-dose interaction where the prior-learning advantage gets smaller as dose increases. H4 will be tested as a difference-in-differences comparison between old targets and novel set C.

We will also use a two-sided equivalence test to ask whether the effort after robust unlearning is practically similar to the never-learned control. More specifically, the robust-unlearning effort ratio will be compared with a preregistered range of 0.90 to 1.10. Growth-curve models of validation likelihood over target-token exposure will give us a second view of the starting level and learning slope. Individual facts and paraphrases will not be treated as independent model replications. The complete training run is the main unit of replication, while item-level analyses will treat facts and topics as nested within each run.

4 Potential Results and Implications

Several outcomes are possible in this experiment, and each one would tell us something different about what remains after unlearning. The table below summarizes the main patterns we would be looking for.

Table 2

Hypothetical outcome patterns and interpretations

Hypothetical result	Interpretation
Positive matched savings after JensUn, equal pre-relearning likelihood, no effect for novel facts, and a decline with intervention dose	Strongest behavioral evidence that earlier target learning leaves a persistent, target-specific effect after robust unlearning
Savings only after the weaker method, together with elevated correct-answer likelihood	Apparent savings is better explained by incomplete or leaky unlearning
Similar speeding up or slowing down for old and novel facts	The unlearning procedure changed general plasticity rather than preserving a target-specific effect
No savings and successful equivalence to naive and rollback controls	Robust unlearning removed any behaviorally detectable relearning advantage within the sensitivity of the study
Relearning is selectively slower for old targets	Unlearning produced target-specific inhibition or damage rather than savings
Savings remains but does not change with intervening training	Earlier learning affects reacquisition, but the result does not look like a graded forgetting pattern

Note. These are hypothetical outcome patterns, not observed data.

The clearest support for H1-H4 would be a pattern where models with prior target learning need fewer target tokens than both the naive-unlearning and rollback controls after JensUn. At the same time, the models should begin relearning with similar target likelihoods, and the same advantage should not appear when they learn novel facts. If the savings gap also becomes gradually smaller as the amount of intervening training increases, the overall behavior would look similar to savings in human memory. Note that this still would not mean that a language model contains the same kind of memory trace as a human. The pattern could come from optimization geometry, distributed representations, or other machine-specific mechanisms, as emphasized by Yu et al. (2026). What it would show is that ordinary retrieval-based unlearning tests may miss a lasting consequence of earlier training.

A more conservative result, and perhaps a more likely one, would be savings after the weaker method but little or no savings after behaviorally robust JensUn. This would support H2 while giving weaker support to H1. If the weaker condition also starts relearning with a higher correct-answer probability, the simpler explanation would be incomplete suppression rather than a hidden savings mechanism. Such a result would fit with Hu et al. (2024), who show that apparently unlearned knowledge can return after additional training, and it would support using both likelihood-based and relearning-based tests when we evaluate unlearning.

A null result would also need some care. A nonsignificant difference by itself would not show that the conditions are equal. However, if the effort ratio falls inside the preregistered equivalence range, target probabilities look like the rollback condition, and general utility is preserved, then we would have evidence that robust unlearning removed a practically meaningful relearning advantage. That result would put a clear boundary on the analogy between human and machine memory. On the other hand, if old targets are relearned more slowly than novel facts, the unlearning process may have created target-specific inhibition or damaged representations needed for the same relation. The novel-fact control is what lets us separate that selective effect from a general loss of learning ability.

The study also matters for how machine unlearning is evaluated in practice. If earlier exposure makes removed information cheaper to reacquire, unlearning benchmarks may need to test more than immediate output suppression. They may also need to ask whether the information stays unusually easy to restore during later, otherwise harmless training. If no target-specific savings remains after strong unlearning, that would instead support a stricter idea of successful removal: not only is the answer unavailable now, but earlier learning also gives no measurable advantage when the model later sees the same information again. Either outcome would encourage cognitive scientists to study models with controlled training histories rather than drawing conclusions only from prompts given to already pretrained systems.

5 Limitations and Possible Improvements

There are several limits to what this study can tell us. First, the main facts are synthetic and are learned through supervised fine-tuning. This gives us a clean experiment, but the result may not generalize to knowledge that is spread across large-scale pretraining. One natural follow-up would be to repeat the basic design with carefully screened pretrained facts and compare the pattern with the procedural dataset.

Second, the result may depend on the model family, model size, whether we use parameter-efficient or full fine-tuning, and the exact unlearning objective. A stronger follow-up would therefore repeat the study across several open-weight model families and scales. This would tell us whether savings is a general feature of relearning after unlearning or something that appears only under a particular training setup.

Third, our intervening-token manipulation should not be read as the same thing as a psychological retention interval. It measures how much unrelated training changes the model between unlearning and relearning. A useful extension would be to vary not only the amount of intervening training but also how related that training is to the target facts. This could show whether interference depends mainly on the number of updates or on what the model learns in between.

Finally, even a close behavioral match with human savings would not establish a shared internal memory process. The present proposal is intentionally a behavioral study. If a savings effect appears, representational analyses could be used in a follow-up to ask what changes inside the model across first learning, unlearning, intervention, and relearning. Those analyses would help explain the machine mechanism without treating a behavioral similarity as proof of a human-like memory system.

6 Conclusion

In this proposal, we use savings in relearning as a controlled behavioral test of what remains after machine unlearning. The main question is whether a language model that learned and then robustly forgot a set of facts still needs less training to learn those same facts again, and whether that possible advantage becomes weaker after more unrelated training. By using fresh procedural facts, resetting optimizer state, measuring target-token exposure to a held-out criterion, separating hallucination from abstention, and comparing prior-learning models with naive-unlearning, sham, rollback, and novel-fact controls, we should be able to tell apart a target-specific effect of earlier learning from contamination, leftover accessibility, or a general change in learning ability. The goal is not to claim that humans and machines remember in the same way. Instead, the study asks whether their behavior can look similar in one specific and measurable part of relearning.

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